

# Timothy Do

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## EDUCATION

### University of California, Irvine

Sep. 2019 – June 2023 (*Expected*)

*B.S. Electrical Engineering, Specialization in Digital Signal Processing*

*Irvine, CA*

- **Cumulative GPA:** 3.912/4.0, **Major GPA:** 3.924/4.0, **Accelerated Status**
- **Relevant Coursework:** Computer Architecture, Control Systems, Digital Image/Signal Processing, Electronics
- **Research Interests:** Hyperspectral Image Processing, Embedded IoT Systems, Parallel GPU Optimizations
- **Clubs and Societies:** IEEE, IPC, Eta Kappa Nu, Tau Beta Pi

## EXPERIENCE

### Industrial Internet of Things Intern

June 2022 – Sep. 2022

*Western Digital*

*San Jose, CA*

- **Skills:** MySQL, Python, Raspberry Pi, Soldering, Linux, Lean Six Sigma
- Certified as a Six Sigma Yellow Belt for learning Six Sigma foundations
- Applied Industrial Toolsets to develop a Sensor Data Analytics Platform
- Saved the company an annual projected cost of \$312,000 from hazard mitigations

### Undergraduate Student Researcher

January 2022 – Present

*Irvine Computer Vision Laboratory at UC Irvine*

*Irvine, CA*

- Researched under Professor Glenn Healey on the topic of Spectral Image Filtering and Baseball Simulations.
- Digitized Imaging Filters into CSVs and compared them in MATLAB against certain chemical emissions.
- Analyzed Hyperspectral Data Cubes in MATLAB and determined dominant bands in SWIR and NWIR spectras

### Vice President

Nov. 2021 – Present

*Institute of Printed Circuits Club at UC Irvine*

*Irvine, CA*

- Reserved club meeting rooms and sent weekly emails to club members regarding club events
- Created the IPC Club Website and managed IPC's social media platforms
- Designed circuit & programming workshops for IPC club members to apply in industrial settings

### I.T. General Assistant

July 2021 – June 2022

*UC Irvine Division of Continuing Education*

*Irvine, CA*

- Helped Set-Up Desktops, Docks, Printers, Webcams, and Computers for over 300+ Employees
- Managed IPs, Subnet Masks, and DNS servers to access local networking services
- Monitored and Updated a Real-Time Web Server (DCE-OLLI) by rebuilding the Frontend/Backend using react.js

## PROJECTS

### The Do-Pro Senior Design Project | *Raspberry Pi, Python, MATLAB, OpenCV*

Sep. 2022 - March 2023

- Led a team of four in creating an embedded stereo-vision camera for ADAS detection and 3D-Reconstruction
- Calibrated the stereo-camera in MATLAB and implemented block matching algorithms with OpenCV
- Designed the button/touch-screen user interface using RPi.GPIO and Tkinter for navigating the camera
- Developed a Power Regression Model for the Do-Pro to perceive depth with 6% error from 1-3 feet

### AM Radio & Amplifier | *LTSpice, Oscilloscopes, Soldering*

April 2022 – May 2022

- Led IPC @ UCI's Two-Part AM Radio Workshop, Creating the Schematic for Optimal Gain and Efficiency
- Designed Audio Amplifier using the *LM741* Op-Amp in an Inverting Configuration with a Single USB 5V Supply
- Designed Envelope Detector using the *1N4148* Diode and RC Band-Pass Filter for Demodulation

### EECS195-Colorization | *Python, MATLAB*

Feb. 2022 – March 2022

- Colorized grayscale images using GAN and Auto-Encoder Neural Networks
- Gathered own datasets by taking pictures of different scenery
- Clustered pixels into superpixel subgraphs for uniform training weights

## SKILLS

**Electronics:** Raspberry Pi, Network Analysis, Cadence, LTSpice, Soldering, Oscilloscope & Multimeter Measurements

**Programming Languages:** Python, Java, C, CUDA, MATLAB, bash, MySQL HTML/CSS/Javascript, VHDL

**Developer Tools:** git, SSH, Makefile, Linux, Azure, VS Code, Visual Studio, PyCharm, Eclipse, Vivado HLS

**Libraries:** OpenCV, Pandas, Numpy, Matplotlib, PIL, Tensorflow, Pytorch, Tkinter RPi.GPIO, Socket, Flask